

Rationalise complex numbers

Basic skills

Q1. a) $z = 2 + 3i$ $z^* = 2 - 3i$

 change sign of imaginary part

b) $z = 4 - i$ $z^* = 4 + i$

c) $z = 2i - 5$ $z^* = -2i - 5$

Q2. $z = 2 + 3i$, $w = 5 - 2i$, $v = 7i - 9$

a) $zw = (2 + 3i)(5 - 2i)$

$$= 10 - 4i + 15i - 6i^2$$
$$= 16 + 11i$$

 $i^2 = -1$

b) $zv = (2 + 3i)(7i - 9)$

$$= 14i + 21i^2 - 18 - 27i$$
$$= -39 - 13i$$

c) $wv^* = (5 - 2i)(-7i - 9)$

$$= -35i + 14i^2 - 45 + 18i$$

$$= -59 - 17i$$

$$\begin{aligned} d) \quad z z^* &= (2+3i)(2-3i) \\ &= 4 + 6i - 6i - 9i^2 \\ &= 4 + 9 = 13 \end{aligned}$$

or note $z z^* = x^2 + y^2$

Q3. By the complex conjugate of denominator on top & bottom

$$\text{so } \frac{4-5i}{4-5i}$$

Competency skills

$$\begin{aligned} \text{Q4. a) } \frac{(3-2i)}{(2+2i)} \times \frac{(2-2i)}{(2-2i)} &= \frac{6-6i-4i+4i^2}{4+4} \\ &= \frac{2-10i}{8} = \frac{1-5i}{4} \end{aligned}$$

use $z z^*$ $+4i-2i$

$$b) \quad \frac{6+5i}{3-2i} \times \frac{3+2i}{3+2i} = \frac{18+12i+15i+10i^2}{9+4}$$

$$= \frac{8 + 27i}{13}$$

$$c) \frac{5(i-6)}{i-1} \times \frac{-i-1}{-i-1} = \frac{5(-i^2 + 6i - i + 6)}{1+1}$$

$$= \frac{5(7+5i)}{2} = \frac{35+25i}{2}$$

$$e) \frac{3i-2}{(3-2i)^2} = \frac{3i-2}{9-12i+4i^2} = \frac{3i-2}{5-12i}$$

$$\frac{3i-2}{5-12i} \times \frac{5+12i}{5+12i} = \frac{15i-10+36i^2-24i}{25+144}$$

$$= \frac{-46-9i}{169}$$

$$f) \frac{(2-i)(i-2)}{5-2i} = \frac{-4-i^2}{5-2i} = \frac{-3}{5-2i}$$

$$\frac{-3}{5-2i} \times \frac{5+2i}{5+2i} = \frac{-15-6i}{25+4} = \frac{-15-6i}{29}$$

Q5. $(3-5i)z = 10+3i$

$$\text{So } z = \frac{10+3i}{3-5i} = \frac{10+3i}{3-5i} \times \frac{3+5i}{3+5i}$$

$$= \frac{30+9i+50i+15i^2}{9+25}$$

$$= \frac{15+59i}{34}$$

Q6. $\frac{1-6i}{3+2i} = \frac{a+bi}{2+5i}$

$$\begin{aligned} \text{So } a+bi &= \frac{(1-6i)(2+5i)}{3+2i} \\ &= \frac{2-12i+5i-30i^2}{3+2i} \end{aligned}$$

$$= \frac{32 - 7i}{3 + 2i} \times \frac{3 - 2i}{3 - 2i}$$

$$= \frac{96 - 21i - 64i + 14i^2}{9 + 4}$$

$$= \frac{82 - 85i}{13}$$

$$\text{So } a = \frac{82}{13} \quad b = -\frac{85}{13}$$

Q7. $(4 + 9i)(z - 3i + 2) = 2 - 5i$

$$z - 3i + 2 = \frac{2 - 5i}{4 + 9i}$$

$$z - 3i + 2 = \frac{2 - 5i}{4 + 9i} \times \frac{4 - 9i}{4 - 9i}$$

$$= \frac{8 - 20i - 18i + 45i^2}{16 + 81}$$

$$z = 3i - 2 + \frac{-37 - 38i}{97}$$

$$z = \left(-2 - \frac{37}{97}\right) + \left(3 - \frac{38}{97}\right)i$$

$$= -\frac{231}{97} + \frac{253}{97}i$$

Q8. $z + z^* = (x + yi) + (x - yi)$
 $= 2x$

$$z - z^* = (x + yi) - (x - yi)$$

$$= 2yi$$

Advanced skills

Q9. To divide z by w i.e. $z \div w$

$$\text{we instead do } \frac{z}{w} = \frac{z}{w} \times \frac{w^*}{w^*} = \frac{zw^*}{|w|^2}$$

$$\text{so } \div w \text{ is same as } \times \frac{w^*}{|w|^2}$$

Q10.

$$\frac{1}{z} + \frac{1}{z^*} = \frac{1}{z} \times \frac{z^*}{z^*} + \frac{1}{z^*} \times \frac{z}{z}$$

$$= \frac{z^*}{zz^*} + \frac{z}{zz^*} = \frac{z^* + z}{zz^*}$$

$$= \frac{z^* + z}{|z|^2} = \frac{2x}{x^2 + y^2}$$

by Q8.
normal rule

Similarly

$$\frac{1}{z} - \frac{1}{z^*} = \frac{z^* - z}{zz^*} = \frac{2yi}{x^2 + y^2}$$

Q11.

$$\frac{a}{2-i} + \frac{bi}{5+2i} = 2+3i$$

↓ could multiply instead

$$\frac{a(2+i)}{4+1} + \frac{bi(5-2i)}{25+4} = 2+3i$$

$$\frac{2}{5}a + \frac{1}{5}ai + \frac{5}{29}bi + \frac{2}{29}b = 2+3i$$

$$\text{So } \frac{2}{5}a + \frac{2}{29}b = 2, \quad \frac{1}{5}a + \frac{5}{29} = 3$$

$$\text{So } \frac{2}{5}a + \frac{10}{29}b = 6$$

$$- \frac{2}{5}a + \frac{2}{29}b = 2$$

$$\frac{8}{29}b = 4$$

$$b = \frac{4 \times 29}{8} = \frac{29}{2}$$

$$\text{So } \frac{2}{5}a + \frac{2}{29} \times \frac{29}{2} = 2$$

$$\frac{2}{5}a + 1 = 2$$

$$a = \frac{5}{2}$$

Q12.

$$z + 2iw = 1$$

$$2z - 8w = 22$$

$$\text{So } 2z + 4iw = 2$$

$$\Rightarrow 4iw + 8w = -20$$

$$(4i+8)w = -20 \quad w = \frac{-20}{8+4i} = \frac{-5}{2+i}$$

$$w = \frac{-5}{2+i} \times \frac{2-i}{2-i} = \frac{-10+5i}{4+1} = -2+i$$

Then

$$z = 1 - 2iw$$

$$= 1 - 2i(-2+i)$$

$$= 1 + 4i - 2i^2 = 3 + 4i$$

Q13.

$zw = 1$ then w is inverse of z

$$(x+yi)(a+bi) = 1$$

$$(xa - yb) + (xb + ya)i = 1$$

$$xa - yb = 1 \quad xb + ya = 0$$

" x, y known" find a, b

$$-ya - \frac{y^2}{x}b = \frac{y}{x}$$

$$ya + xb = 0$$

$$-(\frac{y^2}{x} + x)b = \frac{y}{x}$$

$$-(y^2 + x^2)b = y$$

$$b = \frac{-y}{x^2 + y^2}$$

$$a = \frac{x}{x^2 + y^2}$$

$$\text{So } w = \frac{x - yi}{x^2 + y^2} = \frac{z^*}{|z|^2}$$

i.e. w is $\frac{1}{z}$

Dividing by z is same as $\times \frac{z^*}{|z|^2}$